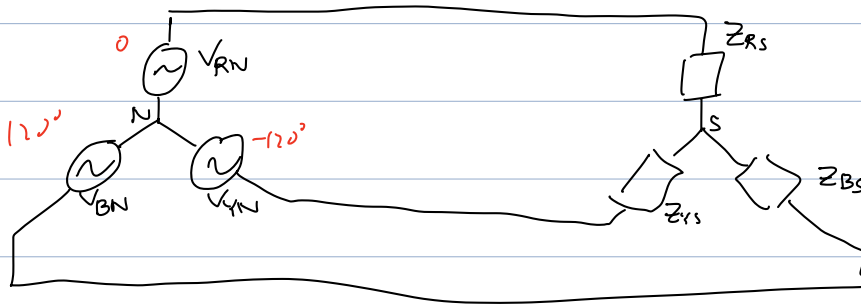


T<sub>2</sub>

Q1. Balanced Star



$$V_{ph} = 11kV \quad Z_{RS} = 10 + j5 = 11.2 \angle 26.6^\circ \Omega$$

$$I_L = \frac{11k}{10 + j5} = 0.88 - j0.44 \text{ kA} = 984 \angle -26.6^\circ \text{ A}$$

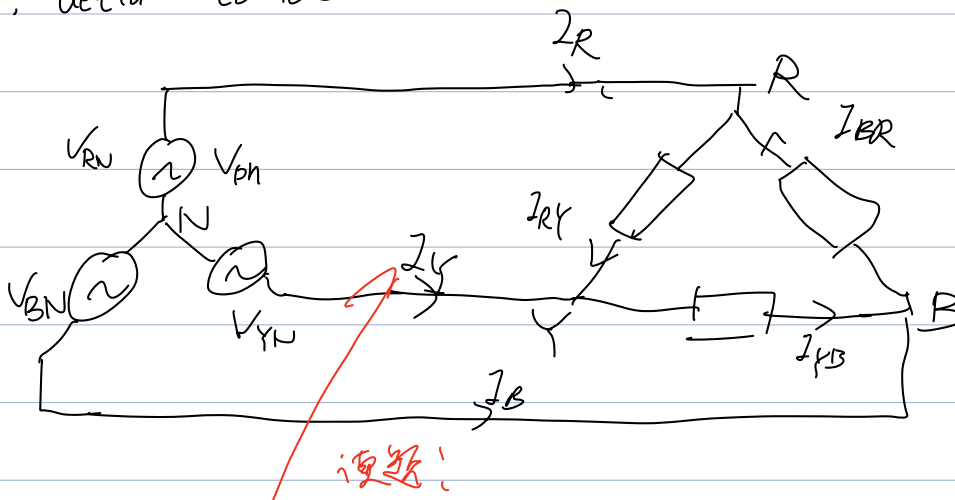
$$P_{total} = 3 |V_{ph}| |I_L| \cos \theta = 29 \text{ MW}$$

$$I_{LR} = \frac{V_{RN}}{Z_{RS}} = \frac{11k \angle 0^\circ}{11.2 \angle 26.6^\circ} = 984 \angle -26.6^\circ \text{ A}$$

$$I_{LY} = \frac{V_{YN}}{Z_{YS}} = \frac{11k \angle -120^\circ}{11.2 \angle 26.6^\circ} = 984 \angle -146.6^\circ \text{ A}$$

$$I_{LB} = \frac{V_{BN}}{Z_{BS}} = \frac{11k \angle 120^\circ}{11.2 \angle 26.6^\circ} = 984 \angle 93.4^\circ \text{ A}$$

Q2. delta connected



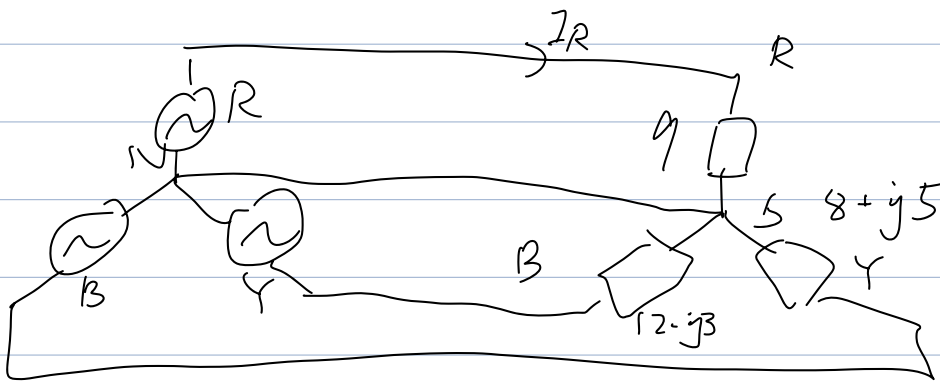
$$Z = 50 + j \cdot 2\pi \cdot 50 \cdot 40m = 50 + 4\pi j = 51.6 \angle 14.1^\circ \Omega$$

$$I_{RY} = \frac{V_{RY}}{Z} = \frac{415 \angle 30^\circ}{51.6 \angle 14.1^\circ} = 8.04 \angle 15.9^\circ A$$

...

$$I_R = \sqrt{3} I_{RY} \angle -30^\circ = 13.9 \angle -14.1^\circ A$$

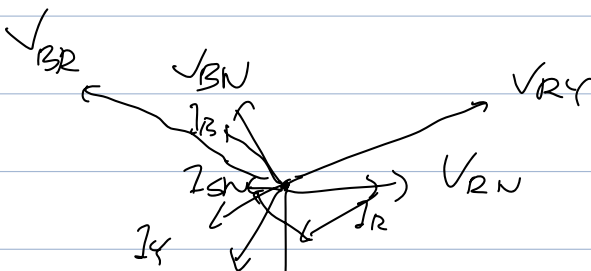
Q3 unbalanced star



$$I_R = \frac{400/\sqrt{3}}{9} = 44.4/\sqrt{3} A$$

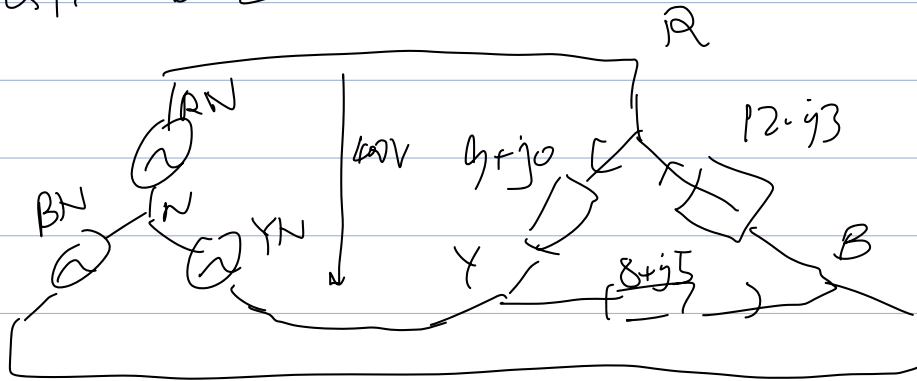
$$I_Y = \frac{400 \angle -120^\circ / \sqrt{3}}{8 + j5} = 42.4 \angle -152^\circ / \sqrt{3} A$$

$$I_B = \frac{400 \angle 120^\circ / \sqrt{3}}{12 - j3} = 32.3 \angle 134^\circ / \sqrt{3} A$$





Q4. unb  $\Delta$



$$I_{RY} = \frac{400 \angle 30^\circ}{9 + j0} = 44.4 \angle 30^\circ \text{ A}$$

$$I_{YB} = \frac{400 \angle -90^\circ}{8 + j5} = 42.4 \angle -122.0^\circ \text{ A}$$

$$I_{BR} = \frac{400 \angle 150^\circ}{12 - j3} = 32.3 \angle 164.0^\circ \text{ A}$$

$$P_{\text{total}} = |V_L| \cdot (|I_{RY}| \cdot \cos 0^\circ + |I_{YB}| \cdot \cos(-32.0^\circ) + |I_{BR}| \cdot \cos(14.0^\circ))$$

$$= 44.68 \text{ kW}$$

Q5.  $|V_L| \angle 30^\circ$

$$I_{RY} = \frac{|V_L| \angle 30^\circ}{Z_{RY}}$$

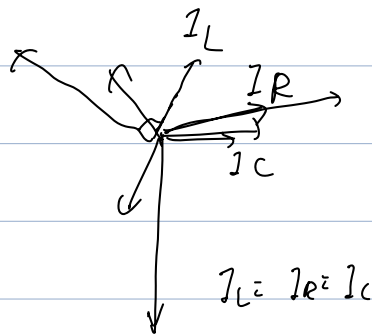
$$I_{YB} = \frac{|V_L| \angle -90^\circ}{Z_{YB}}$$

$$I_{BR} = \frac{|V_L| \angle 150^\circ}{Z_{BR}}$$

$$I_R = I_{RY} - I_{BR} = |V_L| \cdot \left( \frac{\angle 30^\circ}{Z_{RY}} - \frac{\angle 150^\circ}{Z_{BR}} \right)$$

$$I_Y = I_{YB} - I_{RY} = |V_L| \cdot \left( \frac{\angle -90^\circ}{Z_{YB}} - \frac{\angle 30^\circ}{Z_{RY}} \right)$$

$$I_B = I_{BR} - I_{YB} = |V_L| \cdot \left( \frac{\angle 150^\circ}{Z_{BR}} - \frac{\angle -90^\circ}{Z_{YB}} \right)$$



$$\frac{\angle 30^\circ}{|Z_R| \angle 0^\circ} - \frac{\angle 150^\circ}{|Z_L| \angle 90^\circ}$$

$$\frac{\angle -90^\circ}{|Z_C| \angle -90^\circ} - \frac{\angle 30^\circ}{|Z_R| \angle 0^\circ}$$

$$\frac{\angle 150^\circ}{|Z_L| \angle 90^\circ} - \frac{\angle -90^\circ}{|Z_C| \angle -90^\circ}$$

